

Total No. of Questions : 6]

SEAT No. :

P 5515

[Total No. of Pages : 1

BE/INSEM/Oct-91
BE.(I.T.)
INFORMATION AND CYBER SECURITY
(2012 Pattern) (Semester - I)

Time : 1 Hour]

[Max. Marks :30

Instructions to the candidates:

- 1) *Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6.*
- 2) *Figures to the right indicate full marks.*
- 3) *Assume suitable data, if necessary.*

- Q1)** a) Explain the terms: Substitution Cipher and Transposition Cipher. [4]
b) Determine the GCD of (24140, 16762) using Euclidean algorithm. [6]

OR

- Q2)** a) What is an Initialization Vector (IV)? What is its significance? [4]
b) Draw and explain the term Kerberos message sequence. [6]

- Q3)** a) Explain how 48 bit key is generated in DES. [4]
b) What do you mean by X.509 standard for digital certificate? [6]

OR

- Q4)** a) What is the important aspect that establishes trust in digital signatures? [4]
b) Draw AES block diagram and explain the steps in detail. [6]

- Q5)** a) Perform encryption and decryption using RSA algorithm for the following: [5]

$p = 5, q = 11, e = 3$ and $m = 9$

- b) Why are some attacks called as passive? Why other attacks are called active? [5]

OR

- Q6)** a) Discuss the key management: Key Generation, Distribution, Updation. [5]
b) Write a short note on: Security Goals, Security Attacks. [5]

